Question Paper Preview

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Change Font Color :	No
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Sw Testing

Section Id :	640653124670
Number of Questions :	19
Section Marks :	100
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531862668 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE TESTING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536034488. ✔ YES

6406536034489. ✖ NO

Question Number : 2 Question Id : 6406531862669 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Which of the following types of testing of a near-finished software product is done by a set of end users outside the development team?

Options :

6406536034490. ✖ System Testing

6406536034491. ✖ Acceptance Testing

6406536034492. ✔ Beta Testing

6406536034493. ✖ Integration Testing

Question Number : 3 Question Id : 6406531862670 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Which of the following options best describes the purpose of scalability tests?

Options :

6406536034494. ✖ To determine if a system protects data and maintains security related functionality.

6406536034495. ✖ To measure the ability of system to keep operating over specified periods of time.

6406536034496. ✔ To verify that a system can handle load upto its engineering limits.

6406536034497. ✖ To determine whether the system can interoperate with other third-party products.

Question Number : 4 Question Id : 6406531862671 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question

Consider a context free grammar G over a finite alphabet $\sum = \{a, b, c\}$. The following are the production rules for G:

$$S \rightarrow bX$$
$$X \rightarrow bXa$$
$$X \rightarrow c$$

Let S be the starting symbol. Which of the following sets represents the language generated by the given grammar?

Options :

6406536034498. ✔ $\{b^{n+1}ca^n | n \geq 0\}$

6406536034499. ✖ $\{b^n c^{n+1} a^n | n \geq 1\}$

6406536034500. ✖ $\{b^{n+1} c^n a^n | n \geq 0\}$

6406536034501. ✖ $\{b^n c^n a^{n+1} | n \geq 1\}$

Question Number : 5 Question Id : 6406531862672 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question

Which type of client-side testing tries to check for issues related to relationships among different parameters of an input?

Options :

6406536034502. ✔ Parameter level bypass testing

6406536034503. ✖ Control flow level bypass testing

6406536034504. ✖ Value level bypass testing

6406536034505. ✖ User-session data based testing

Question Number : 6 Question Id : 6406531862673 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

The following table lists some categories of entities on the world wide web.

Category	Description
1. Static Web Page	A. This type of web page is unvarying and the same to all users.
2. Dynamic Web Page	B. It is a collection of web pages and associated software elements that are related semantically by content and syntactically through links.
3. Web Sites	C. This type of web page is created by a program on demand.
4. Web Services	D. It is a system of software that allows machines to interact with each other through a network.

Match each category with the correct description.

Options :

6406536034506. ✖ 1 - B, 2 - C, 3 - A, 4 - D

6406536034507. ✖ 1 - A, 2 - D, 3 - B, 4 - C

6406536034508. ✔ 1 - A, 2 - C, 3 - B, 4 - D

6406536034509. ✖ 1 - B, 2 - A, 3 - D, 4 - C

Question Number : 7 Question Id : 6406531862674 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider that a user makes an online purchase by making payment through a payment gateway, after the payment is successful, the user is sent back to the e-commerce website with an order confirmation. Which type of transition would take place in the Application Transition Graph (ATG) in this scenario?

Options :

6406536034510. ✖ Form Link Transition

6406536034511. ✔ Redirect Transition

6406536034512. ✖ Operational Transition

6406536034513. ✖ Simple Link Transition

Question Number : 8 Question Id : 6406531862675 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question

Consider ISP with four blocks as $[A_1, A_2, A_3, A_4]$, $[B_1, B_2]$, $[C_1, C_2, C_3]$, and $[D_1, D_2]$. The base choice is (A_1, B_2, C_3, D_2).

What is the number of *total* test cases to satisfy base choice coverage criterion?

Options :

6406536034514. ✖ 6

6406536034515. ✖ 7

6406536034516. ✔ 8

6406536034517. ✖ 9

Question Number : 9 Question Id : 6406531862676 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question
Consider the following Java program.

```java
class Item {
  protected String itemName;
  protected double itemPrice;
  protected int stockQuantity;
  public Item() { itemName = ""; itemPrice = 0.0; stockQuantity = 0; }
  public void setItemName(String name) { this.itemName = name; }
  public void setItemPrice(double price) { this.itemPrice = price; }
  public void setStockQuantity(int quantity) { this.stockQuantity = quantity; }
  public void displayStockInfo() {
    System.out.println(itemName + ", " + stockQuantity + " units");
  }
}
class Product extends Item {
  public void setStockQuantity(int quantity) {
    super.setStockQuantity(quantity);
  }
}
class Laptop extends Product {
  protected int stockQuantityUnits;
  public void setStockQuantity(int quantity) {
    stockQuantityUnits = quantity + 10;
  }
}
class Inventory {
  public static void main(String[] args) {
    Laptop laptop = new Laptop();
    phone.setItemName("Laptop 11");
    phone.setItemPrice(36000);
    phone.setStockQuantity(25);
    phone.displayStockInfo();
  }
}
```

The program outputs Laptop 11, 0 units, instead of Laptop 11, 35 units.
It is because the inherited method Laptop::setStockQuantity does not call Product::setStockQuantity and therefore does not define the instance variable stockQuantity in Item.
Which of the following types of anomalies or faults are present in the given scenario?

Options :

6406536034518. ✖ State Visibility Anomaly (SVA)

6406536034519. ✔ State Definition Anomaly (SDA)

6406536034520. ✖ State Definition Inconsistency Anomaly (SDIA)

6406536034521. ✖ Inconsistent Type Use (ITU)

Question Number : 10 Question Id : 6406531862677 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following Java program.

```java
class X {
    public void c() { System.out.println("Method c from class X"); }
    public void d() { System.out.println("Method d from class X"); }
    public void e() { System.out.println("Method e from class X"); }
}
class Y extends X {
    public void d() { System.out.println("Method d from class Y"); }
    public void f() { System.out.println("Method f from class Y"); }
}
class Z extends Y {
    public void c() { System.out.println("Method c from class Z"); }
    public void f() { System.out.println("Method f from class Z"); }
}
public class Test {
    public static void main(String[] args) {
        Z obj = new Z();
        obj.c();
        obj.d();
        obj.e();
        obj.f();
    }
}
```

Which of the following sets of methods will be invoked in the given program?

Options :

6406536034522. ✖ {X :: c(), Y :: d(), X :: e(), Y :: f()}

6406536034523. ✖ {X :: c(), X :: d(), X :: e(), Z :: f()}

6406536034524. ✔ {Z :: c(), Y :: d(), X :: e(), Z :: f()}

6406536034525. ✖ {Z :: c(), X :: d(), X :: e(), Z :: f()}

Question Number : 11 Question Id : 6406531862678 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question
Consider the following code statement (ground string) and a mutant:

```
int x = x % 2; // Ground String
```

```
int x = x / 2; // Mutant-1
```

Which of the following types of mutation operators can be applied to the ground string to obtain Mutant-1?

Options :
6406536034526. ✖ Assignment operator replacement
6406536034527. ✖ Absolute value insertion
6406536034528. ✖ Scalar variable replacement
6406536034529. ✔ Arithmetic operator replacement

Question Number : 12 Question Id : 6406531862679 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following code segment.

```
1  class Metric {
2      protected double value;
3      public Metric() {
4          this.value = 0;
5      }
6      public Metric(double value) {
7          this.value = value;
8      }
9      public void print() {
10         System.out.println("Metric Value: " + value);
11     }
12 }
13 class Precision extends Metric {
14     public Precision(double val) {
15         super(value); // LINE-15
16     }
17 }
```

Consider that a mutant is created by deleting the statement super(value); in LINE-15. Which of the following mutation operators is used to create the mutant?

Options :

6406536034530. ✖ Hiding Variable Insertion (HVI)

6406536034531. ✔ Parent Constructor Deletion (PCD)

6406536034532. ✖ Overriding Method Deletion (OMD)

6406536034533. ✖ Super Keyword Deletion (SKD)

Question Number : 13 Question Id : 6406531862680 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question
Consider the following code segment of a Java servlet. The atomic sections are marked as

	/* meter_readings stores the electricity consumption in kilowatt-hours in a manufacturing facility over the past year. */ ArrayList<Double> meter_readings = null; response.setContentType("text/html"); PrintWriter out=response.getWriter();

P_1	out.println ("<HTML><HEAD><TITLE>Report</TITLE></HEAD><BODY>");
	String facilityCode = request.getParameter("facilityCode");
	/* fetchMeterData() takes the facility code (facilityCode) as input, runs a query in the database, and returns a ArrayList object containing the electricity consumption for the past year at this facility. */
	meter_readings = fetchMeterData(facilityCode);
	double total_consumption = 0;
	double avg_consumption = 0.0;

	if(meter_readings == null) {
P_2	out.println("Invalid Facility Code ");
	} else {
	for (Double c : meter_readings) {
P_3	total_consumption += c;
	}
P_4	avg_consumption = total_consumption / meter_readings.size();
	if(avg_consumption <= 24000.0)
P_5	out.println("Consumption is within normal range. ");
	else
P_6	out.println("Consumption exceeds the limit. ");
	}
P_7	out.println ("</BODY></HTML>"); out.close();

$P_1, P_2, P_3, \cdots, P_7$.

Identify the component expression that corresponds to the given code segment.

Options :

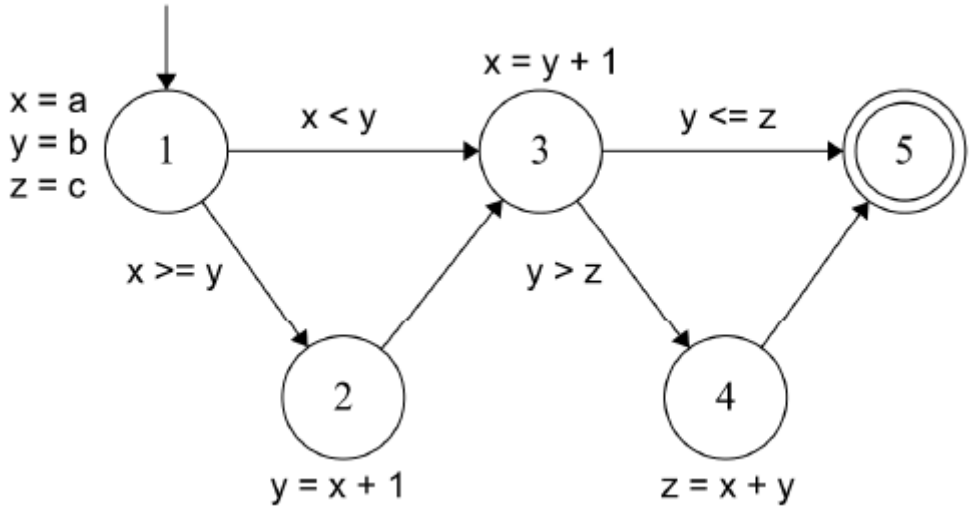
6406536034534. ✗ $P_1 \cdot ((P_2^* \cdot P_3 \cdot (P_4|P_5))|P_6) \cdot P_7$

6406536034535. ✓ $P_1 \cdot (P_2|(P_3^* \cdot P_4 \cdot (P_5|P_6))) \cdot P_7$

6406536034536. ✖ $P_1 \cdot (P_2|(P_3 \cdot P_4^* \cdot (P_5|P_6))) \cdot P_7$

6406536034537. ✖ $P_1 \cdot (P_2|(P_3^* \cdot (P_4 \cdot P_5|P_6))) \cdot P_7$

Question Number : 14 Question Id : 6406531862681 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question
Consider the following data flow graph.



Given a set of test paths $\{[1,3,5],[1,2,3,5],[1,3,4,5]\}$, which of the following coverage criteria is satisfied for the variable x?

Options :
6406536034538. ✖ All-Defs Coverage only
6406536034539. ✔ All-Uses Coverage only
6406536034540. ✖ All-DU-Paths Coverage
6406536034541. ✖ None of these

Question Number : 15 Question Id : 6406531862682 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following code segment for symbolic execution.

```
boolean isPositive(int n) { return n > 0; }
void verify(int x) {
    int m = (x + 2) * (x - 2);
    if(!isPositive(x)) {
        if(m <= 20) {
          abort(); // LINE 1
        }
    }
}
```

Assume that x_0 respresents the symbolic value for the variable x in the method void verify(int x). Which of the following path constraints would reach the abort() call at LINE-1?

Options :

6406536034542. ✖ $(x_0 > 0) \wedge ((x_0 * x_0) - 2 > 20)$

6406536034543. ✖ $(x_0 \geq 0) \wedge ((x_0 * x_0) - 4 > 20)$

6406536034544. ✔ $(x_0 \leq 0) \wedge ((x_0 * x_0) - 4 \leq 20)$

6406536034545. ✖ $(x_0 < 0) \wedge ((x_0 * x_0) - 4 > 20)$

Question Number : 16 Question Id : 6406531862683 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider test-driven development (TDD) of a software program that works with geometric objects. The implementation for the method TriangleType::type should identify the type of the triangle based on the given side lengths.

The following is a test case for this method.

```
public class TestTriangleType {
    private TriangleType obj;

    @Test
    public void test() {
        obj = new TriangleType();
        assertEquals(obj.type(3, 4, 5), "SCALENE");
        assertEquals(obj.type(5, 5, 5), "EQUILATERAL");
        assertEquals(obj.type(5, 5, 8), "ISOSCELES");
        assertEquals(obj.type(0, 0, 0), "INVALID");
        assertEquals(obj.type(1, 2, 3), "INVALID");
        assertEquals(obj.type(0, 5, 5), "INVALID");
        assertEquals(obj.type(-1, 5, 5), "INVALID");
    }
}
```

Which of the following are correct implementations of the method?

Options :

6406536034546. ✖

```
public class TriangleType {
    public String type(int a, int b, int c) {
        if (a + b <= c || a + c <= b || b + c <= a) { return "INVALID"; }
        if (a == b && b == c) { return "EQUILATERAL"; }
        if (a == b || a == c || b == c) { return "ISOSCELES"; }
        return "SCALENE";
    }
}
```

6406536034547. ✔

```java
public class TriangleType {
    public String type(int a, int b, int c) {
        if (a <= 0 || b <= 0 || c <= 0) { return "INVALID"; }
        if (a + b <= c || a + c <= b || b + c <= a) { return "INVALID"; }
        if (a == b && b == c) { return "EQUILATERAL"; }
        if (a == b || a == c || b == c) { return "ISOSCELES"; }
        return "SCALENE";
    }
}
```

6406536034548. ✖

```java
public class TriangleType {
    public String type(int a, int b, int c) {
        if (a <= 0 || b <= 0 || c <= 0) { return "INVALID"; }
        if (a + b <= c || a + c <= b || b + c <= a) { return "INVALID"; }
        if (a == b || a == c || b == c) { return "ISOSCELES"; }
        return "SCALENE";
    }
}
```

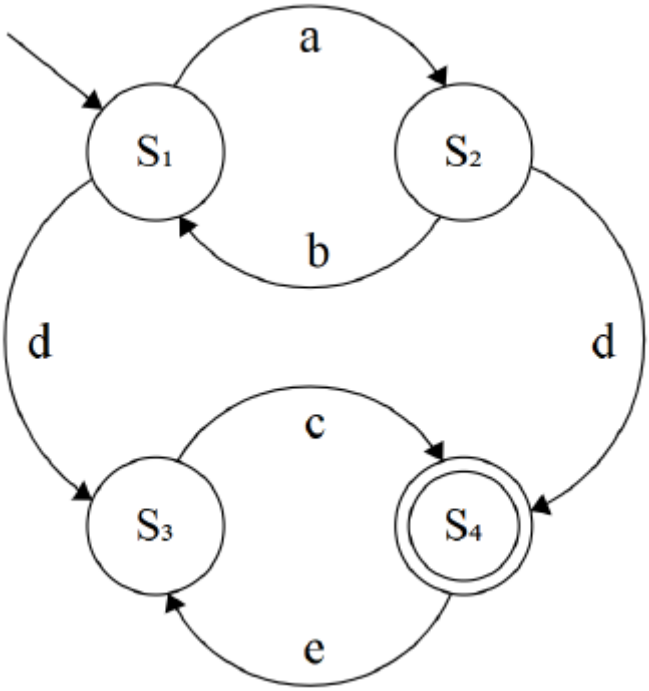
6406536034549. ✔

```java
public class TriangleType {
    public String type(int a, int b, int c) {
        if (a <= 0 || b <= 0 || c <= 0) { return "INVALID" };
        if (!(a + b > c && a + c > b && b + c > a)) { return "INVALID"; }

        int uniqueSides = 3;
        if (a == b && b == c) { uniqueSides = 1; }
        else if (a == b || a == c || b == c) { uniqueSides = 2; }

        switch (uniqueSides) {
            case 1: return "EQUILATERAL";
            case 2: return "ISOSCELES";
            default: return "SCALENE";
        }
    }
}
```

Question Number : 17 Question Id : 6406531862684 Question Type : MSQ
Correct Marks : 5 Max. Selectable Options : 0
Question Label : Multiple Select Question
Consider the following finite state machine (FSM).



Which of the following strings can be accepted by the given FSM?

Options :
6406536034550. ✖ abed
6406536034551. ✔ adec
6406536034552. ✖ abde
6406536034553. ✔ ad

Question Id : 6406531862685 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (18 to 19)

Question Label : Comprehension

Consider the predicate $p = (a \wedge b) \vee \neg c$ and its truth table given below.
Note that the blank cells represent the value F in the truth table.

#	a	b	c	p	p_a	p_b	p_c
1	T	T	T	T	T	T	
2	T	T		T			
3	T		T			T	T
4	T			T			T
5		T	T		T		T
6		T		T			T
7			T				T
8				T			T

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 18 Question Id : 6406531862686 Question Type : MCQ Correct Marks : 5

Question Label : Multiple Choice Question

Which of the following is the set of RACC pairs for literal b?

Options :

6406536034554. ✖ $\{(1,3),(2,6),(3,7)\}$

6406536034555. ✖ $\{(1,5)\}$

6406536034556. ✖ $\{(2,6),(3,7),(4,8)\}$

6406536034557. ✔ $\{(1,3)\}$

Question Number : 19 Question Id : 6406531862687 Question Type : MCQ Correct Marks : 5

Question Label : Multiple Choice Question

Which of the following is the set of RICC pairs for literal b?

Options :

6406536034558. ✖ $\{(1,5),(2,6),(3,7)\}$ for

$p = T$, no feasible pair for $p = F$

6406536034559. ✖ $\{(1,5),(3,7)\}$ for $p = T$. and $\{(5,7)\}$ for $p = F$

6406536034560. ✔ $\{(2,4)\}$ for $p = T$, and $\{(5,7)\}$ for $p = F$

6406536034561. ✖ $\{(2,4),(5,7)\}$ for $p = T$, and $\{(3,7)\}$ for $p = F$

Question Id : 6406531862688 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (20 to 21)

Question Label : Comprehension

Consider the control flow graph (CFG), $G = \{V, E\}$, where

- Set of vertices $V = \{1, 2, 3, 4, 5, 6, 7, 8\}$
- Set of edges $E = \{(1, 2), (1, 3), (2, 4), (3, 4), (4, 5), (4, 7), (5, 6), (5, 7), (6, 8), (7, 8)\}$
- Initial vertex $V_0 = 1$
- Final vertex $V_f = 8$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 20 Question Id : 6406531862689 Question Type : MCQ Correct Marks : 5

Question Label : Multiple Choice Question

What is the cyclomatic complexity of G?

Options :

6406536034562. ✖ 3

6406536034563. ✔ 4

6406536034564. ✖ 5

6406536034565. ✖ 6

Question Number : 21 Question Id : 6406531862690 Question Type : MCQ Correct Marks : 5

Question Label : Multiple Choice Question

How many test requirements are there for edge-pair coverage on G?

Options :

6406536034566. ✖ 9

6406536034567. ✖ 10

6406536034568. ✔ 11

6406536034569. ✖ 12